Madhan Ravi

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SUMMARY

Python backend and MERN stack developer with hands-on experience building, testing, and deploying full-stack applications. Builds REST APIs with Django REST Framework and Express.js, real-time features with Socket.io, and JWT authentication; writes automated tests (Jest, Supertest, pytest) with GitHub Actions CI, and deploys live on Vercel, Render, Neon, and MongoDB Atlas.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL, Java, HTML/CSS
Frontend: React.js, Next.js, Tailwind CSS, Vite, React Router, Context API, Axios
Backend: Node.js, Express.js, Socket.io (WebSockets), Django, Django REST Framework, PyQt5
Databases: MongoDB (Mongoose, Atlas), PostgreSQL (Neon), SQLite, Redis (Upstash)
Testing & CI/CD: Jest, Supertest, pytest, GitHub Actions, Postman
Cloud & DevOps: Vercel, Render, Docker, Docker Compose
Tools & Practices: Git, GitHub, Swagger/OpenAPI, JWT Auth, bcrypt, REST API Design, CORS, Rate Limiting
Core Concepts: OOP, Data Structures & Algorithms, CRUD Architecture, Role-Based Access Control

EXPERIENCE

Full Stack Developer Intern *July 2026 - Present*
Marzelet Info Technology Pvt Ltd *Chennai, Tamil Nadu*

- Built and shipped production-facing UI components using React.js, Next.js, TypeScript, and Tailwind CSS, working directly within a live codebase under senior engineer review.
- Designed and implemented RESTful APIs using Python, Django, and Django REST Framework to connect frontend and backend systems for live product features.
- Queried, modeled, and persisted application data in PostgreSQL, contributing to schema design decisions for new features.
- Debugged issues spanning frontend, backend, and API layers, improving reliability of production-grade code before release.
- Collaborated using Git-based version control and team code-review workflows, following industry-standard software engineering practices.

PROJECTS

Chatz - Real-Time Chat App *September - October 2026*
React, Node.js, Express.js, Socket.io, MongoDB, JWT, Jest, GitHub Actions | GitHub | Live Demo

- Built a real-time chat app with Socket.io where authenticated users create rooms and message instantly, with room-based broadcasting; messages persist in MongoDB across 3 Mongoose models (User, Room, Message) and reload in order with sender usernames.
- Secured the app by verifying a JWT in the Socket.io handshake and protecting all 3 REST route groups (auth, rooms, history) with JWT middleware; passwords hashed with bcrypt.
- Wrote 12 automated tests (Jest, Supertest) for the auth, room, and message-history endpoints against an isolated test database; GitHub Actions CI runs them on every push and pull request in under a minute.
- Deployed live across 3 platforms (React on Vercel, Socket.io server on Render, MongoDB Atlas), resolving production CORS and SPA-routing issues.

Django Job Portal *August - September 2026*
Python, Django, DRF, PostgreSQL (Neon), Docker, pytest, GitHub Actions | GitHub | Live Demo

- Eliminated duplicate and unauthorized job applications by enforcing database-level uniqueness constraints and ownership checks, across a role-based portal spanning 4 core Django apps (profiles, postings, applications, resumes).
- Secured and scaled the API layer by implementing JWT authentication and role-based permissions across all endpoints, adding pagination, search, filtering, and ordering; validated every endpoint using Postman before release.
- Enabled one-command environment setup and consistent deployments by containerizing the full stack with Docker and Docker Compose, and shipped interactive Swagger/OpenAPI + ReDoc API documentation; later made the project publicly testable end-to-end by deploying to Vercel with a Neon serverless PostgreSQL database.
- Wrote 39 pytest tests covering JWT auth, role-based permissions, ownership checks, input validation, and the duplicate-application constraint, run by GitHub Actions CI on every push; the tests caught and fixed a 500 error on partial job updates.

EDUCATION

Bachelor of Computer Application *Oct 2021 - May 2024*
Sindhi College of Arts and Science | CGPA: 7.8/10 *Chennai, Tamil Nadu*